

Approximation of Portfolio Kurtosis through Sum of Squared Quadratic Forms

Dany Cajas*

[First version: June 2023] [Latest version: June 2023]

Abstract

This work presents a new representation of portfolio kurtosis through the sum of squared quadratic forms that allows us to pose a convex portfolio optimization model for an approximate kurtosis that has high accuracy and give us a high improvement in speed calculation respect to the exact parametric portfolio kurtosis model. This new model is based on second order cone and semidefinite programming, and it has the flexibility to pose problems like kurtosis constraints, kurtosis minimization, risk adjusted return ratio maximization and risk parity based on kurtosis. Also, it can be easily extended to the heuristic of lower semi kurtosis.

Keywords: finance, convex programming, portfolio optimization, kurtosis, risk parity.

JEL Codes: C61, G11

Acknowledgement: I want to thanks and dedicate this work to my parents Daniel and Pilar, and thanks to Catherine, Mappy and Alberto for their support.

*E-mail: dcajasn@gmail.com

1 Introduction

The optimization of portfolio kurtosis using the disciplined convex programming (DCP) methodology¹ was proposed by Cajas (2022). However, this model does not scale too well when we increase the number of assets, making this model suitable for medium scale problems but not for large scale problems. To overcome this drawback, this work proposes a new formulation of portfolio kurtosis through sum of squared quadratic forms. Based on this new formulation, we built a new convex portfolio model that optimize a close approximation of portfolio kurtosis. This formulation is based on second order cone and semidefinite programming. We are going to run some numerical examples using the approximate kurtosis model using Python 3.9 ², CVXPY³ and MOSEK⁴ solver; in order to show the properties of the approximate kurtosis portfolios compared to kurtosis portfolios. Finally, we are going to compare the performance of the new model respect to the exact parametric portfolio kurtosis model.

2 Kurtosis as a Sum of Squared Quadratic Forms

Cajas (2022) proposed a quadratic form formula for portfolio kurtosis as follows:

$$\mu_4 = (x \otimes x)' \Sigma_4 (x \otimes x) \quad (1)$$

where Σ_4 is the square cokurtosis matrix of size $n^2 \times n^2$ and x are the portfolio weights of n assets. The cokurtosis square matrix can be decomposed as the sum of n^2 matrixes A_i that has a Kronecker square root matrix B_i , this means that:

$$\Sigma_4 = \sum_{i=1}^{n^2} (B_i \otimes B_i) \quad (2)$$

To get the value of B_i , we need to use the rearrangement operator or block vectorization operator $\text{vec}^{m,n}$ defined by Loan and Pitsianis (1992) and use the property of block vectorization operator $\text{vec}^{n,m}(C + D) = \text{vec}^{n,m}(C) + \text{vec}^{n,m}(D)$. Using this property we can split the block vectorization of cokurtosis matrix Σ_4 into the sum of block vectorization of matrixes A_i . Then using the necessary and sufficient conditions

¹Grant and Boyd (2006)

²<https://anaconda.org>

³Diamond and Boyd (2016) and Agrawal et al. (2018)

⁴ApS (2022)

proposed in Ojeda (2015)⁵, we can obtain the Kronecker square root matrix B_i from a matrix A_i using the spectral decomposition of matrix $\text{vec}^{n,n}(\Sigma_4)$ as follows:

$$\begin{aligned}
 \text{vec}^{n,n}(\Sigma_4) &= \sum_{i=1}^{n^2} \text{vec}^{n,n}(A_i) \\
 \text{vec}^{n,n}(\Sigma_4) &= \sum_{i=1}^{n^2} \text{vec}(B_i) \text{vec}(B_i)' \\
 \text{vec}^{n,n}(\Sigma_4) &= \sum_{i=1}^{n^2} \lambda_i q_i q_i' \\
 \text{vec}(B_i) &= \sqrt{\lambda_i} q_i \\
 B_i &= \text{vec}^{-1}(\sqrt{\lambda_i} q_i) \\
 B_i &= (\text{vec}(I_n)' \otimes I_n) (I_n \otimes \sqrt{\lambda_i} q_i)
 \end{aligned} \tag{3}$$

where λ_i and q_i are the eigenvalue and eigenvector i of matrix $\text{vec}^{n,n}(\Sigma_4)$. If we replace 2 in 1, we get the following expression:

$$\begin{aligned}
 \mu_4 &= (x \otimes x)' \Sigma_4 (x \otimes x) \\
 \mu_4 &= (x \otimes x)' \left(\sum_{i=1}^{n^2} (B_i \otimes B_i) \right) (x \otimes x) \\
 \mu_4 &= \sum_{i=1}^{n^2} (x' \otimes x') (B_i \otimes B_i) (x \otimes x) \\
 \mu_4 &= \sum_{i=1}^{n^2} (x' B_i \otimes x' B_i) (x \otimes x) \\
 \mu_4 &= \sum_{i=1}^{n^2} (x' B_i x \otimes x' B_i x) \\
 \mu_4 &= \sum_{i=1}^{n^2} (x' B_i x)^2
 \end{aligned} \tag{4}$$

We can see that the portfolio kurtosis can be expressed as the sum of n^2 squared quadratic forms, this formulation can be used to approximate kurtosis using the first K matrixes B_i .

⁵A matrix A has a Kronecker square root B , $A = B \otimes B$, only if $\text{vec}^{n,m}(A)$ is symmetric and has rank 1. Also, $\text{vec}^{n,m}(A) = \text{vec}(B) \text{vec}(B)'$.

$$\mu_4 \approx \sum_{i=1}^K (x' B_i x)^2 \quad (5)$$

This approach can be easily extended to lower semi kurtosis just replacing the cokurtosis square matrix with the lower semi cokurtosis matrix.

3 Convex Optimization of Approximate Kurtosis

We can approximate the minimization of kurtosis using the following nonconvex model:

$$\begin{aligned} \min_x \quad & \sum_{i=1}^K (x' B_i x)^2 \\ \text{s.t.} \quad & \sum_{i=1}^n x_i = 1 \\ & x \geq 0 \end{aligned} \quad (6)$$

In the case all B_i matrixes where positive semidefinite, we can transform the non-convex model into a convex model using the second order cone as follow:

$$\begin{aligned} \min_x \quad & \|g\|_4 \\ \text{s.t.} \quad & \left\| B_i^{1/2} x \right\|_2 \leq g_i \quad \forall i = 1, \dots, K \\ & \sum_{i=1}^n x_i = 1 \\ & x \geq 0 \end{aligned} \quad (7)$$

However, in most cases matrixes B_i are indefinite, so we need to use semidefinite programming. First, we need to remember that $x' C x = \text{tr}(C x x')$, where $\text{tr}(\cdot)$ is the trace operator. So equation 5 can be expressed as:

$$\mu_4 \approx \sum_{i=1}^K \text{Tr}(B_i x x')^2 \quad (8)$$

If we use an auxiliary variable X , that $X \succeq x x'$, we can approximate the optimization model 6 into a semidefinite programming problem:

$$\begin{aligned}
& \min_x \quad \|g\|_2 \\
& \text{s.t.} \quad \begin{bmatrix} X & x \\ x' & 1 \end{bmatrix} \succeq 0 \\
& \quad g_i = \text{Tr}(B_i X) \quad \forall i = 1, \dots, K \\
& \quad \sum_{i=1}^n x_i = 1 \\
& \quad x \geq 0 \\
& \quad X = X'
\end{aligned} \tag{9}$$

4 Numerical Examples

We select 30 assets (i.e., stocks JCI, TGT, CMCSA, CPB, MO, T, APA, MMC, JPM, ZION, PSA, BAX, BMY, AAPL, PCAR, BA, TMO, TXT, DE, MSFT, HPQ, SEE, VZ, CNP, NI, JNJ, PFE, AMZN, GE and GOOG) from the S&P 500 (NYSE) and download daily adjusted closed prices from Yahoo Finance for the period from January 1, 2016 to December 30, 2021. Then, we calculated daily returns building a returns matrix of size $T = 1508$ and $N = 30$. To calculate the portfolios we use Python 3.9, CVXPY and MOSEK solver.

4.1 Minimization of Approximate Kurtosis

In this section, we are going to compare two portfolios: minimum kurtosis, variance and approximate kurtosis.

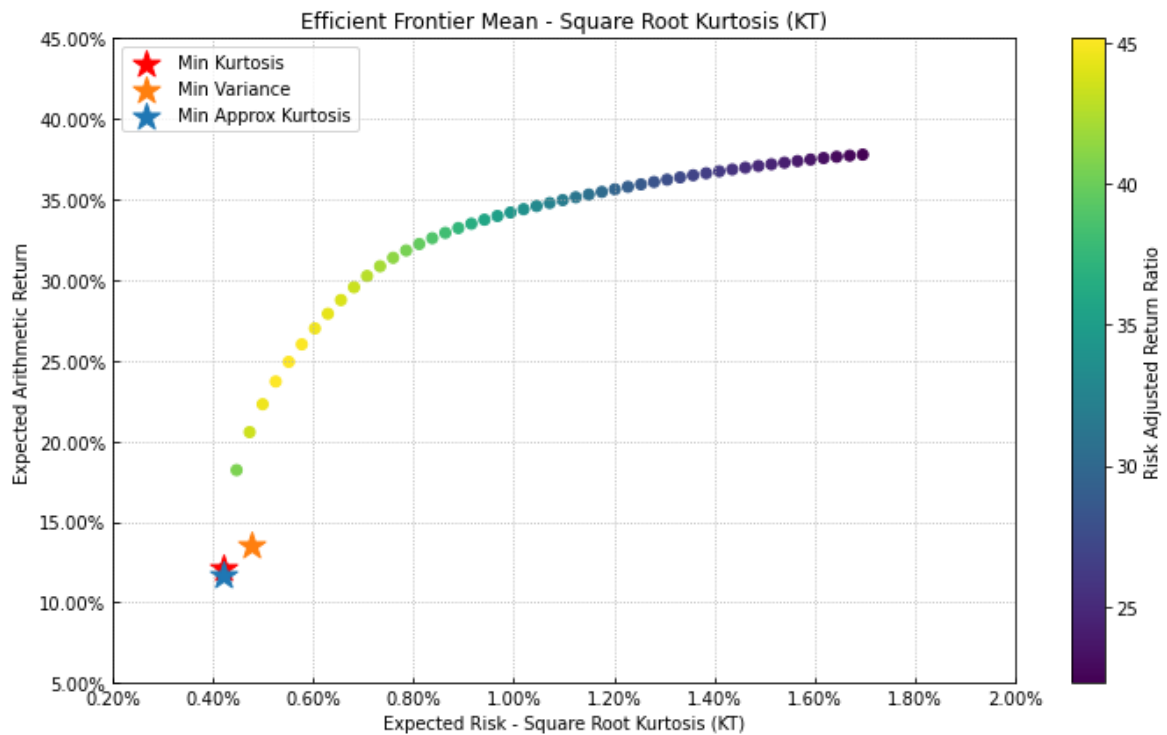


Figure 1: Optimal Portfolios in Efficient Frontier

In figure 1 we can see that kurtosis and approximate kurtosis portfolios are very close, and both are far away from variance portfolio. This means that the approximate kurtosis model is a good approximation to kurtosis portfolio for the minimization of risk objective function.

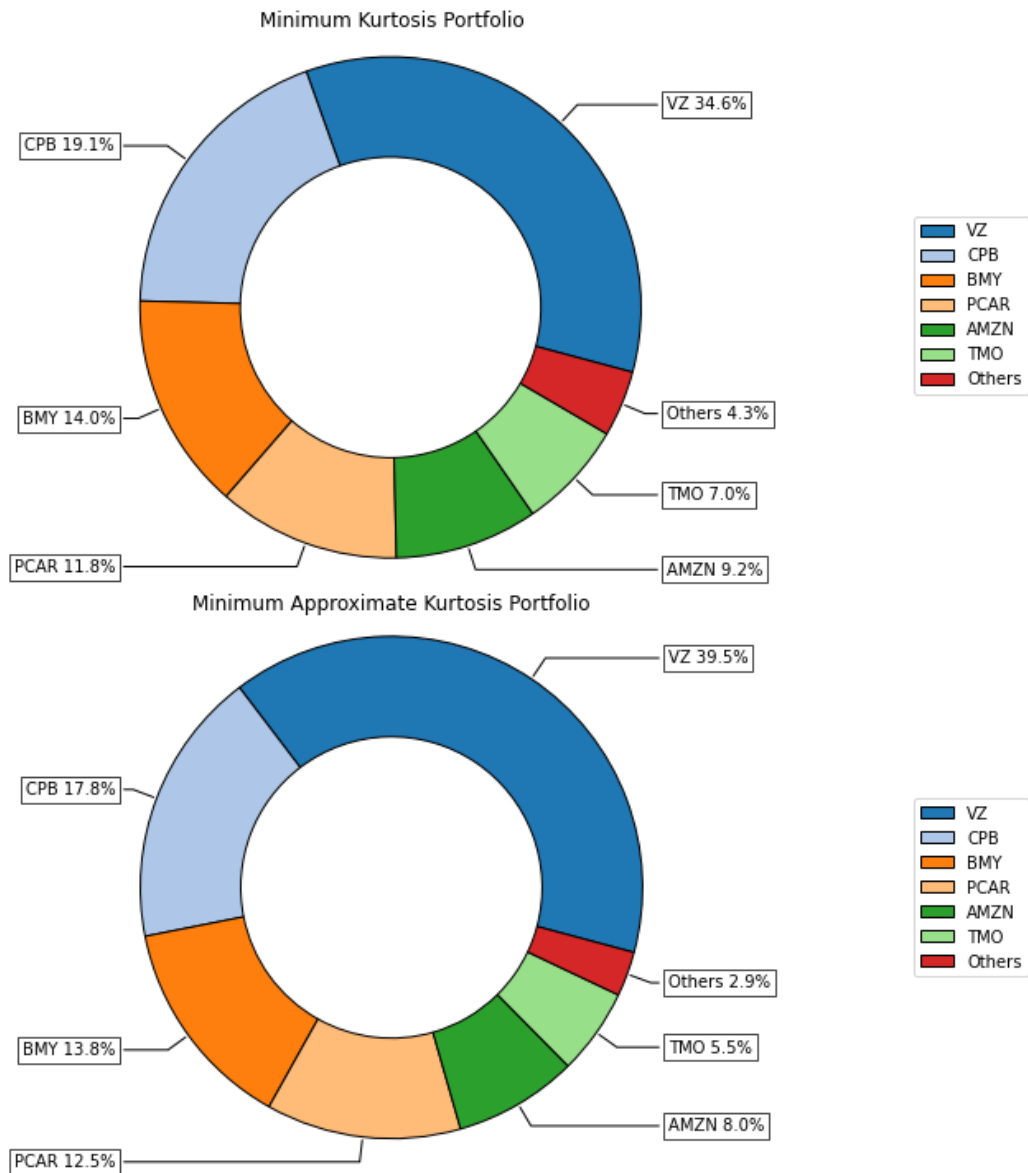


Figure 2: Optimal Portfolios Composition

In figure 2 we can see that both portfolios have similar compositions.

4.2 Maximization of Return Approximate Kurtosis Ratio

In this section, we are going to compare two portfolios: max risk adjusted return ratio for kurtosis, variance and approximate kurtosis.

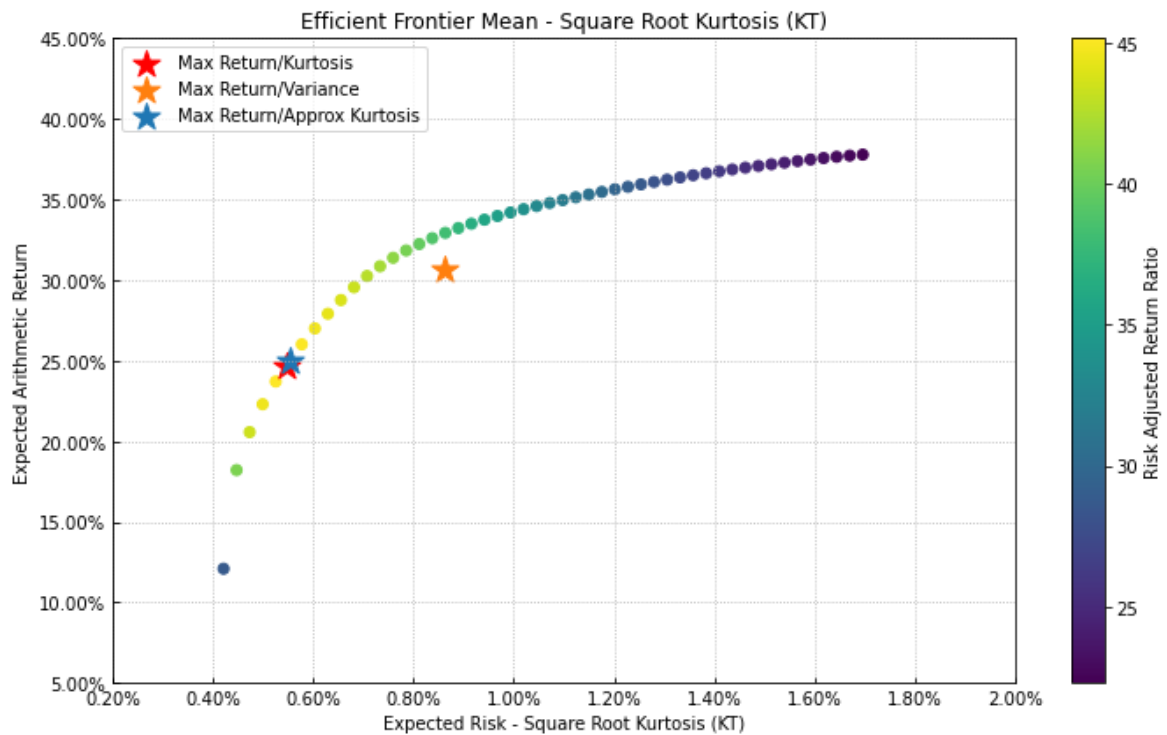


Figure 3: Optimal Portfolios in Efficient Frontier

In figure 3 we can see that kurtosis and approximate kurtosis portfolios are very close, and both are far away from variance portfolio. This means that the approximate kurtosis model is a good approximation to kurtosis portfolio for the risk adjusted return ratio objective function.

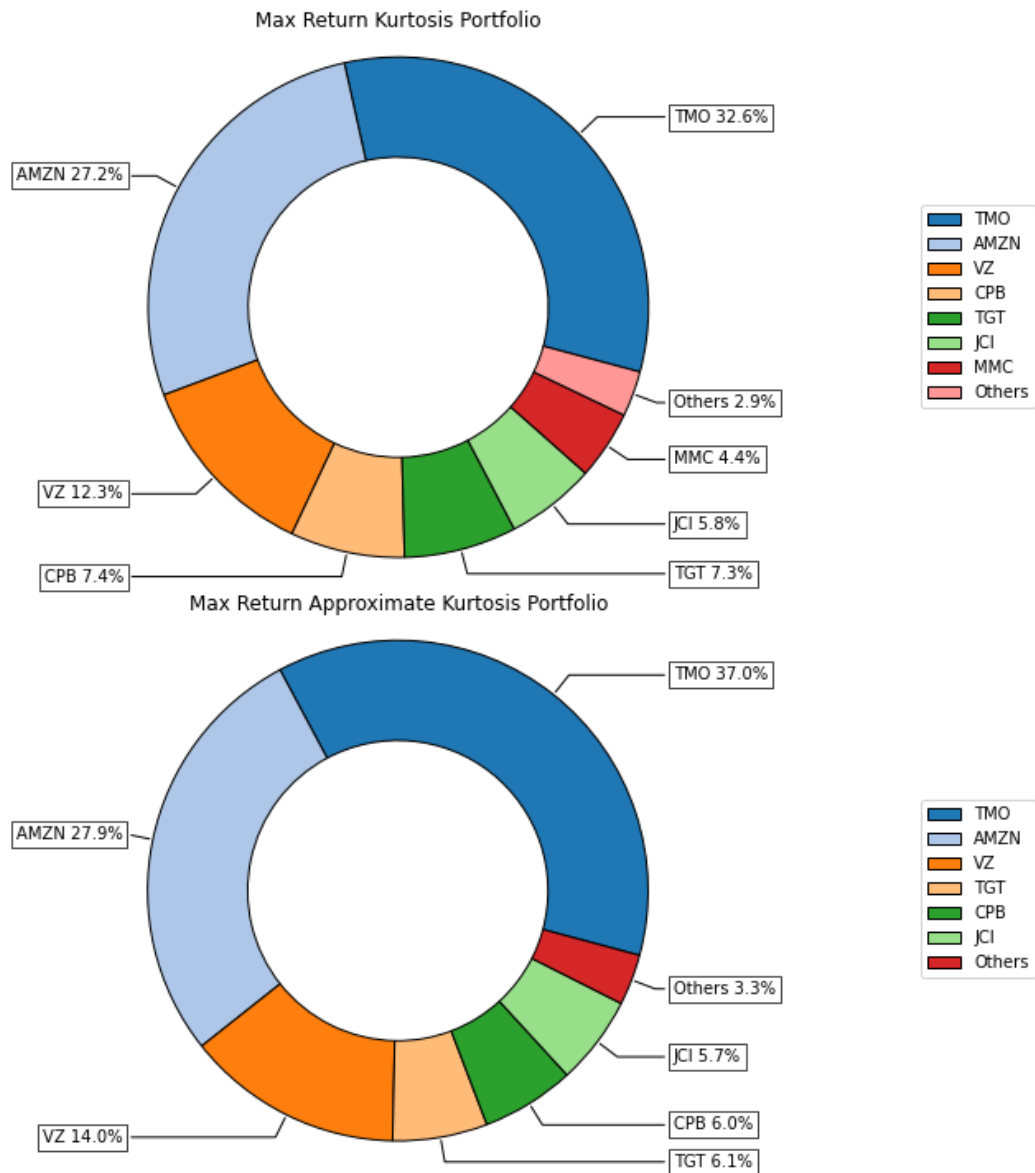


Figure 4: Optimal Portfolios Composition

In figure 4 we can see that both portfolios have similar compositions.

4.3 Risk Parity Based on Approximate Kurtosis

In this section, we are going to compare two portfolios: risk parity kurtosis and approximate kurtosis.

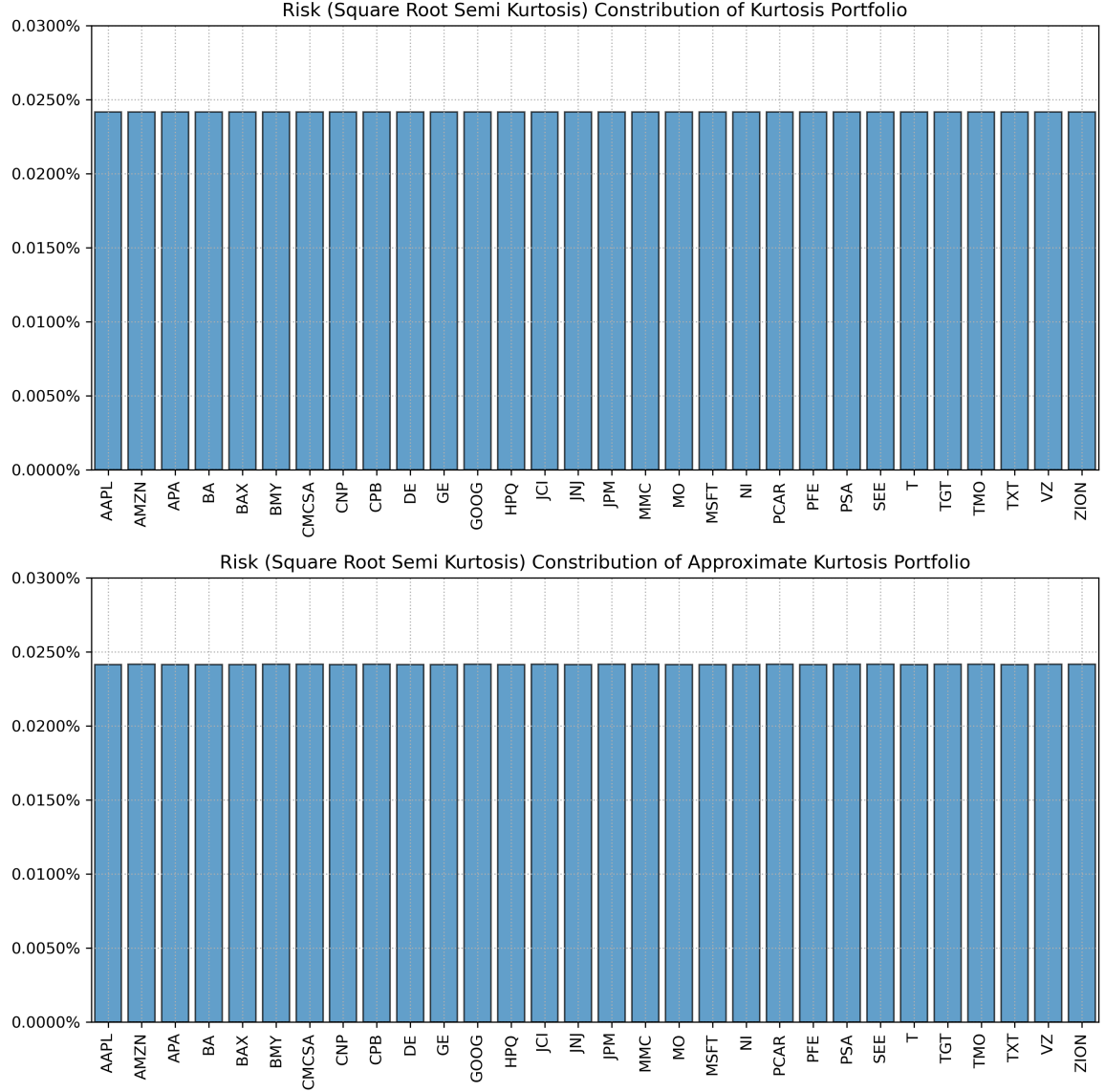


Figure 5: Risk Contribution for Optimal Portfolios

In figure 5, we can see that both portfolios effectively split the fourth root of kurtosis among all assets. This means that the approximate kurtosis model is a good approximation to kurtosis portfolio for the risk parity objective function.

5 Performance of Kurtosis Portfolios

In this section we compare the performance of parametric model proposed by Cajas (2022) and approximate kurtosis model taking $K = 2N$, this means that we are going to use the first $2N$ matrixes B_i to make the approximation of kurtosis. We simulated 10,000 and 20,000 normal returns for 10, 50, 80 and 100 assets. Then, we estimated

3 instances of the portfolios that maximizes three objective functions: minimization of risk, maximization of risk adjusted return and risk parity for kurtosis. Then, we computed the average time and average value for each combination of sample size, objective function and risk measure. The simulation was computed using a computer with a M1 MAX processor and 64 GB of RAM, Python 3.9, CVXPY package and MOSEK solver.

Objective	Sample Size	Assets	Parametric (1)	Approximate (2)	(1)-(2)	(1)/(2)
MinRisk	10000	10	0.03	0.04	-0.01	0.75
		50	3.66	1.03	2.63	3.55
		80	43.4	6	37.4	7.23
		100	152.02	17.94	134.08	8.47
	20000	10	0.03	0.04	-0.01	0.75
		50	3.67	1.03	2.64	3.56
		80	40.66	6.01	34.65	6.77
		100	149.77	17.47	132.3	8.57
Risk Adjusted Return Ratio	10000	10	0.04	0.05	-0.01	0.80
		50	3.5	0.97	2.53	3.61
		80	39.59	6.41	33.18	6.18
		100	139.21	15.43	123.78	9.02
	20000	10	0.03	0.04	-0.01	0.75
		50	3.4	0.94	2.46	3.62
		80	38.51	5.37	33.14	7.17
		100	145.11	15.5	129.61	9.36
Risk Parity	10000	10	0.04	0.05	-0.01	0.80
		50	4.35	1.17	3.18	3.72
		80	45.47	7.75	37.72	5.87
		100	162.75	22.28	140.47	7.30
	20000	10	0.04	0.05	-0.01	0.80
		50	4.28	1.15	3.13	3.72
		80	45.5	7.72	37.78	5.89
		100	162.94	23.55	139.39	6.92

Table 1: Average Run Time in Seconds of Kurtosis Portfolios

We can see in Table 1 that average run time of approximate kurtosis model are lower than scenario kurtosis for samples higher than 10 assets. The difference is too high in terms of speed, in the best case with 100 assets the approximate model is 9.36 times faster than the parametric kurtosis models. This improvement in speed makes the approximate kurtosis model more suitable for medium size and large scale problems.

We can see in Table 2 that average objective value of parametric model and approximate model are the same for minimization of risk and risk parity objective functions. In the case of maximization of risk adjusted return, we can see that the difference of both ratios is very small. These results suggest that the approximate model is a good

Objective	Sample Size	Assets	Parametric (1)	Approximate (2)	(1)-(2)
MinRisk	10000	10	0.0011	0.0012	-0.0001
		50	0.0033	0.0033	0
		80	0.005	0.005	0
		100	0.0068	0.0068	0
	20000	10	0.0013	0.0013	0
		50	0.0033	0.0033	0
		80	0.0057	0.0057	0
		100	0.0069	0.0069	0
Risk Adjusted Return Ratio	10000	10	1.0238	1.0183	0.0055
		50	0.3615	0.3607	0.0008
		80	0.3326	0.3302	0.0024
		100	0.1627	0.1623	0.0004
	20000	10	0.8229	0.8179	0.005
		50	0.3269	0.3265	0.0004
		80	0.3407	0.3398	0.0009
		100	0.249	0.2482	0.0008
Risk Parity	10000	10	0.0015	0.0015	0
		50	0.0049	0.0049	0
		80	0.0079	0.0079	0
		100	0.0102	0.0102	0
	20000	10	0.0018	0.0018	0
		50	0.0052	0.0052	0
		80	0.0084	0.0084	0
		100	0.01	0.01	0

Table 2: Average Objective Value of Kurtosis Portfolios

convex alternative in terms of speed and accuracy respect to the parametric model.

6 Conclusions

The proposed convex formulation of portfolio approximate kurtosis allows us to optimize kurtosis with a high accuracy and speed compared to the convex formulation for parametric kurtosis proposed by [Cajas \(2022\)](#). This new formulation has the flexibility to allow us to pose several problems like kurtosis constraints, kurtosis minimization, risk adjusted return ratio maximization and risk parity based on kurtosis. This model relies in a semidefinite constraint and a second order cone constraint and it scales better than parametric kurtosis formulation for medium scale problems. Another advantage is that this formulation can be solved using disciplined convex programming softwares that supports second order cone programming and semidefinite programming. In addition, this approximate formulation allows us to use the heuristic of lower semi kurtosis using a similar approximation for lower semi cokurtosis matrix, that minimizes the left

tail of portfolio returns distribution. Finally, this formulation makes portfolio optimization based on kurtosis accessible to most users because it requires less computational resources and can be solved using state of art solvers.

References

- Agrawal, A., R. Verschueren, S. Diamond, and S. Boyd (2018). A rewriting system for convex optimization problems. *Journal of Control and Decision* 5(1), 42–60.
- ApS, M. (2022). *MOSEK Optimizer API for Python 10.0.38*.
- Cajas, D. (2022, 8). Convex optimization of portfolio kurtosis. *SSRN Electronic Journal*.
- Diamond, S. and S. Boyd (2016). CVXPY: A Python-embedded modeling language for convex optimization. *Journal of Machine Learning Research* 17(83), 1–5.
- Grant, M. and S. Boyd (2006). Disciplined convex programming. In L. Liberti and N. Maculan (Eds.), *Global optimization : from theory to implementation*, Nonconvex Optimization and its Applications, pp. 155–210. Springer. https://web.stanford.edu/~boyd/papers/disc_cvx_prog.html.
- Loan, C. V. and N. Pitsianis (1992). Approximation with kronecker products.
- Ojeda, I. (2015). Kronecker square roots and the block vec matrix. *The American Mathematical Monthly* 122(1), 60.